

Cyclistic Bikes - Data Analysis Project Report

Ask

Business Task Statement

The business task we will solve is how do annual members use Cyclistic bikes differently from casual riders; and subsequently, how do we convert casual riders into annual members?

Our stakeholder is Lily Moreno, Director of Marketing at Cyclistic, and will oversee the implementation of the recommended strategy.

Prepare

Data Sources

Where is the data located? For the purpose of this exercise, the data has been sourced from Divvy Bikes, Chicago, all data has been sourced from this link:

<https://divvy-tripdata.s3.amazonaws.com/index.html>

The data is now stored locally on my laptop, and on Google Cloud, for analysis in Google BigQuery. The aggregated data is also stored on Tableau Public for the purpose of creating visualisations.

Data organisation: Data collected from April 2020 onwards is organised into monthly packets. (Before April 2020 the data was organised into quarters, but this data does not form part of this analysis.)

Bias / Credibility of the data: the data is considered reliable as it has been collected by the company in question, not a third party. The data is original because it has not been tampered with before any processing within this analysis. The data is comprehensive because it covers a broad period of time, and no data has been omitted during capture. The data is as current as possible, considering we want to analyse a whole year of data. And the data can clearly be cited as collected by the company whom it is about.

Licensing: The data has been used under this licence:

<https://divvybikes.com/data-license-agreement>

Privacy: The licence states that data must be anonymised. All personally identifiable information such as credit card details have already been removed.

Security: The raw data is open source, so there are no security concerns for this project.

Integrity: Integrity of the data can be confirmed as it was sourced internally to the company which it is about (for the purpose of this exercise). I question the reliability of a small number of records, for example those with duplicate ride-IDs, or rides which appear to end before they begin. The duplicate records will be disregarded, to avoid skewing the analysis, these are due to a crossover of recording data between the May and June data sets. We have also been made aware that rides which appear to end before they start are due to bikes being taken out of service for maintenance; these records will also be disregarded. Outliers of extremely short and long ride duration will also be discarded during cleaning, the threshold of this operation is discussed in detail during this report.

Answering the Question: The data will allow the question to be answered because I can separate Annual Members from Casual Users and look at how their behaviour tends to differ. I will then recommend a strategy to convert Casuals to Members.

Timeframe: The data timeframe we will be looking at is January - December 2024 only. No other period of time will be considered.

Code:

The SQL code used for cleaning and analysis can be found in a text file alongside this report.

Process

Cleaning and Manipulation of the Data

Tools: To clean and analyse the data, I have used SQL with Google BigQuery. Later, I have completed an add-on to this project, in DuckDB / DBeaver, to avoid the complication of signing up to BigQuery again. SQL on either database system allows me to combine all months of data, and perform analysis on it as a whole, which would not be possible with the processing power of an application such as Microsoft Excel.

Integrity: The data's integrity has been ensured by ensuring the total number of records in the combined dataset is equal to the sum of the monthly datasets, minus the number of records we have stated will be dropped, this has been verified in the SQL code.

Cleaning: As part of the cleaning process, I have:

- Combined all months of data into one dataset, and excluded duplicate Ride-IDs, as we are aware between May and June some rides were recorded into both monthly datasets.
- Calculated the length of each ride, and added a column for this information.
- Calculated the day of the week that each ride took place on, and added a column for this information.
- Converted the day of the week format to: 1 = Monday, 2 = Tuesday... 6 = Saturday, 7 = Sunday, so that a more user friendly visualisation will be produced.
- Create a new column called weekday_weekend, and populate this with either "weekday" or "weekend", according to the above day of the week column.
- Create an "hour of the day" column, based on the start time of the ride.
- Deleted records where the ride length is less than or equal to 0, ie where the end time of the ride is before the start time, as we have been made aware that these records mean the bike was out of service for maintenance, so we do not want these records to form part of our analysis.
- Removed ride length outliers that skew the average by large amounts.
 - After comparing mean with median and percentile measures, I found ride lengths were strongly right-skewed. While casual users' mean ride length was 25.1 minutes versus 12.8 for members, the medians were much closer (12.1 vs 8.7), indicating that a small number of very long rides inflated the average. To mitigate this, I removed ride length outliers, as detailed below.
 - Upper outlier threshold - I tested a range of different percentiles and the length of ride that corresponds. It was decided that 120 mins was a sensible upper outlier threshold, as it balanced business plausibility, and statistical integrity. A two hour ride is a plausible occurrence for those on a leisure ride, while this figure represents the 99th+ percentile of data.
 - Lower outlier threshold - from a business plausibility perspective a 30 second trip seems possible. Rides under 30 seconds were found to account for 1.6% of the data, which is not negligible. Therefore the distribution of this data was explored; the distribution was found to be relatively smooth, so it was clear that there were no logging artefacts. All things considered, I decided it was best to discard rides under 30 seconds from the analysis; this is because by the time a customer has hired a bike, ridden a very short journey to save a slightly longer walk, then off-hired the

bike, then 30 seconds is the lowest length of ride that I would consider to be plausible from a business perspective. I am not envisaging issues with the data here, I am classing rides under 30 seconds as abandoned trips, false starts, mistaken hires etc that do not represent typical customer behavior.

- As a final comparison - I calculated high level statistics both with and without outliers removed. This was found to have a relatively large impact on mean, but a relatively low impact on median and percentiles, which is a good sign that the chosen outlier thresholds are appropriate.
- Records with NULL start station names / IDs were investigated. These were found not to be as a result of a data import error, but as a result of rides which have started at a location away from a specific docking station. All records with a missing start station name, were also missing start station ID, and vice versa. All records did however contain start latitude and longitude coordinates, which describe the start location precisely. As a result of this investigation, these records were not dropped. The same tests were performed on end station name and end station ID fields, the results were the same, so no action was required.
- All other fields were checked for further NULL values, none were found, so no action was required in this regard.

We know that the data has now been cleaned because:

- The number of records in the cleaned data set is equal to the sum of the records in the 12 monthly datasets, less the sum of all records that were dropped, as detailed above.
(5,860,357 (original import with duplicates already dropped) - 1600 (rides with duration 0 or less) - 133709 (rides with outlier durations) = 5725048 (number of records included on final cleaned dataset))
- The ride lengths are now all positive values.
- Outliers have been removed, according to appropriate thresholds: the data now contains no records of rides less than 30 seconds or greater than 2 hours.
- The day of the week value is now always an integer between 1-7. Spot checks on different days of the week have been carried out, against an alternative source, and these were found to be correct
- Rideable_type field has been checked and it contains one of three values: electric_bike, classic_bike, or electric_scooter. Field format is string (or the equivalent VARCHAR in DuckDB).
- Member_casual field has been checked and it contains one of two values, member or casual. Field format is string (equivalent VARCHAR in DuckDB).
- Month of ride, which is extracted from the started_at field, and is always an integer between 1-12.
- Weekday_weekend field is always either "weekday" or "weekend", no other values are present in this field.
- Hour of the day column contains only numerical values between 0-23.

Analyse

Organisation of Data: the data has now been organised and key fields have been identified:

- Day_of_week
- Rideable_type
- Member_casual
- Ride_length_in_mins
- Started_at
- Ended_at
- Weekday_weekend

- Hour_of_day
- Start_station_name
- End_station_name

Summary of Analysis

Trends / Relationships: The visualisations produced as part of this project, included below in this report, are recommended for showing the trends most effectively. However I will provide a summary here too.

Overview

- Members took 76% more rides than Casuals.
- Casuals' rides were on average 56% longer than Members' rides.
- As a result, 13% more minutes of ride time were hired out to Casuals than Members, meaning Casuals represent a customer-base that is essential to retain.

Hour of the Day

- Member ride frequency peaks between 5pm-6pm (at ~388k), and very high usage also occurs between 3pm-4pm (~248k), and 4pm-5pm (~338k). A morning peak can also be seen in the data between 8am-9am (at ~252k).
- Casual ride frequency shows a similar but less exaggerated trend, with a much lower peak of ~198k between 5pm-6pm.
- This pattern suggests potential commuting usage, mainly by members, but somewhat also by casuals.
- Average length of Casual journeys dramatically increase between 8am-11am, then gradually reduce throughout the rest of the day. While Members' journey times are much less varied, with no major pattern. This pattern in Casual journey length suggests leisure usage (typically slower journeys) which begins in the morning and gradually tapers off as the day goes on.

Day of the Week / Weekends

- Members: When normalised per day, members exhibit higher weekday ride frequency, consistent with commuter-style usage patterns.
- Casuals: Casuals show higher weekend ride frequency and longer average weekend ride durations, suggesting leisure-oriented behaviour. However, both groups demonstrate meaningful activity across the full week, indicating somewhat blended usage patterns rather than purely commuter or leisure segmentation.
- The average length of Members' rides stays fairly consistent throughout the week with a slight increase at the weekend, suggesting a small amount of Member-leisure usage at weekends.
- Casuals showing longer weekday journey times than Members suggests potential mid-week leisure usage.
- Total ridden minutes: During the week, Members ride for more minutes in total per day than Casuals; while the reverse is true at weekends.

Month of the Year

- Number of rides: Both Members and Casuals take substantially more rides in the summer months, with a sharp decline from October to December as colder weather sets in. The number of rides re-increases from January - September. Members take more rides than Casuals, in all months of the year.
- Total length of all rides: a similar trend exists for the total length of all rides; the summer sees a huge increase, while the winter sees a huge drop. The summer sees Casuals overtake Members in this metric, and the winter sees Members re-overtake Casuals. This

potentially indicates that leisure usage makes up more ridden minutes than commuting usage from June - September only, while the reverse is more likely for the rest of the year.

- Average length of each ride: Casuals exhibit longer ride lengths each month as the summer approaches, this declines again as we move into winter to levels similar to that of Members who tend to make quicker journeys all year round. Members exhibit a similar trend, but this is far less pronounced. This again points towards suggestions that leisure-associated behaviour (longer journeys), take place more in the summer, which would be unsurprising due to the warmer weather. Surprisingly, October slightly bucks the trend for Casuals, showing a small increase on an otherwise downward trend into winter. This could be attributed to the increased rainfall in October in Chicago, with less experienced riders riding slower than usual due to the wet weather.

Rideable Type

- Classic bikes are used by Members 86% more frequently than Casuals, however Casuals' average ride lengths are 81% longer, resulting in the total ride time for Casuals and members being very similar (~3% more for Members)
- Electric Bikes are also used more frequently by Members than Casuals (77% more), while the average ride length for E-Bikes is only 33% more for Casuals.
- This means that electric bikes are ridden for 32% more minutes by Members, than Casuals. This could indicate that electric bikes are a popular choice for commuters due to their slightly faster journey times.
- Electric Scooters are used substantially less than the other two rideable types (they account for just 1.82% of all ridden minutes), within that segment of the data, there are 45% more rides on electric scooters by Casuals, than Members. This could be attributed to Casuals desire for a novelty method of transport for enjoyment during leisure usage.
- Casuals on average also take 41% longer journeys on e-scooters than Members, resulting in the total number of minutes hired out to Casuals being 104% more than to Members.
- Casuals consistently take longer journeys across all rideable types, which could point towards less confidence as riders than Members, something which Cyclistic could help improve to convert Casuals to Members.
- Overall, Classic Bikes account for 54.06% of all minutes hired out, E-Bikes account for 44.11% of minutes hired out, and E-scooters account for 1.82% of minutes hired out.

Starting / Ending Locations (no visualisations)

- Docking Station usage: Aggregation 10 shows the number and percentage of journeys (split by members and casuals), where the journey is started or ended at a docking station, as opposed to starting at an undocked street location. This aggregation shows that a similar percentage of journeys (80-83%) involved stations for Members as did Casuals. Meaning approximately 17-20% of journeys started at non-docking stations. While this is an interesting statistic, there is no real Member/Casual behavioural difference, so there is no real insight to gain here that is of relevance to the brief.
- Diversity of locations: Aggregation 11 shows the number of unique start and end locations in the data, split by members and casuals. For Members, there were 1752 unique starting locations, and 1755 unique ending locations; for Casuals, there were 1638 unique starting locations, and 1672 unique ending locations. This shows ~6% more locations were used by Members than Casuals, which may represent commuting tendencies, with workplaces being in a wider variety of locations than tourist attractions which leisure users are more likely to visit. This aggregation naturally excludes non-docking station locations, as only start_station_name and end_station_name is considered, which (as discussed during cleaning) excludes rides that were to or from non-docking stations.
- Top 10 starting locations: Aggregation 12 and 13 shows the top 10 start and end locations for members and casuals. The data for Casuals shows locations such as Millennium Park, Shedd Aquarium and Theater on the Lake as within the top10 locations; while for Members,

Kingsbury St & Kinzie St, Clinton St & Washington Blvd, and Clinton St & Madison St are within the top 10 locations. This pattern is consistent with leisure-oriented use for casual riders and more routine, potentially commuter-style usage for members. Again, this aggregation naturally excludes non-docking station locations, as only the fields “start_station_name” and “end_station_name” are considered, which (as discussed during cleaning) excludes rides that were to or from non-docking stations.

With more time and resources, I would look to investigate trip distance (not just duration) for casuals, to calculate average speeds of travel, compared to members, to see if that supports existing theories.

How does this analysis answer the business question:

Business question 1: How do annual members use Cyclistic bikes differently from casual riders?

- Casuals tend to take fewer, longer journeys than Members; and Members take more, shorter journeys. Casuals account for 13% more ridden minutes than Members.
- Weekends are the most popular part of the week for Casuals, while weekdays are the most popular part of the week for Members.
- I would suggest this shows that Casuals tend to use Cyclistic for Leisure purposes where they take their time riding and use the service more at weekends, while Members tend to use the service in a way that is consistent with commuting behaviour, as they tend to use the service more during the week, and tend to be in more of a hurry.
- Both Members and Casuals use Cyclistic more in the summer than the winter. This trend is more exaggerated for Casuals, resulting in Casuals overtaking Members on ridden minutes in the summer months. I suggest that this is due to Members and Casuals preferring to use traditional public transport in the winter, to keep warm and dry; and I would suggest this is more prevalent for Casuals (likely majority tourists) because Members (likely majority professionals) often don't have a choice but to get to work quickly, whereas tourists are more likely to be in a position to take their time and choose.
- Looking at Rideable type: Casuals' average journey lengths are longer on all rideable types, however, this is particularly the case on Classic Bikes, I suggest this is because Casuals tend to be less experienced riders so cannot cycle as fast as Members.
- Members have shorter average journey lengths, which suggests they may tend to be more experienced riders. The trend also exists for E-bikes and E-scooters, however by nowhere near as much, likely because the speed of E-bikes and E-scooters is determined less by the experience level of the rider.
- E-scooters are more popular among Casuals, this could be explained by Casuals seeing E-scooters as a novelty mode of transport; Casuals may be in the city for leisure purposes and want to experience something new, whereas Members may want a faster and more reliable mode of transport such as a Classic Bike or E-bike.
- Starting / Ending locations: the data shows that all Casuals traveled to fewer locations than all Members; this could be explained as Casuals tending to visit a smaller number of leisure destinations, rather than a large number of potential work places. The top 10 locations for Members and Casuals reinforces this theory.

Business Question 2 - How do we convert casual riders into annual members?

- This business question is answered in the “Act” section at the end of this report.

Share

Data Visualisations

Most of the below visualisations show 3 key metrics. These metrics are:

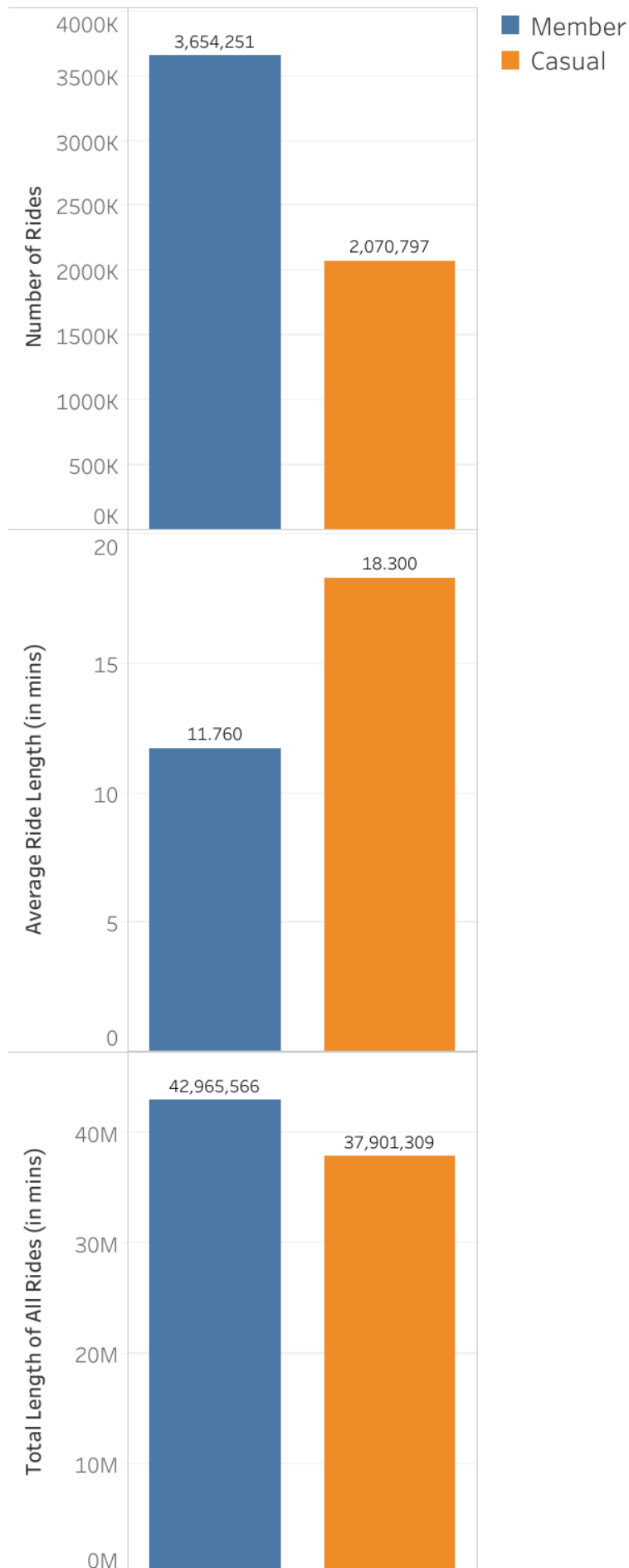
- 1 - number of rides
- 2 - average length of ride
- 3 - total length of all rides

Data is always broken down by customer type, as this analysis is all about Members and Casuals behaviour.

I have created 6 visualisations, shown below, showing Member and Casual behaviour, across the following metrics:

- 1 - Summary - includes all data, unfiltered by time or rideable type.
- 2 - Hour of the day
- 3 - Day of the week
- 4 - Month
- 5 - Weekday / Weekend
- 6 - Rideable type

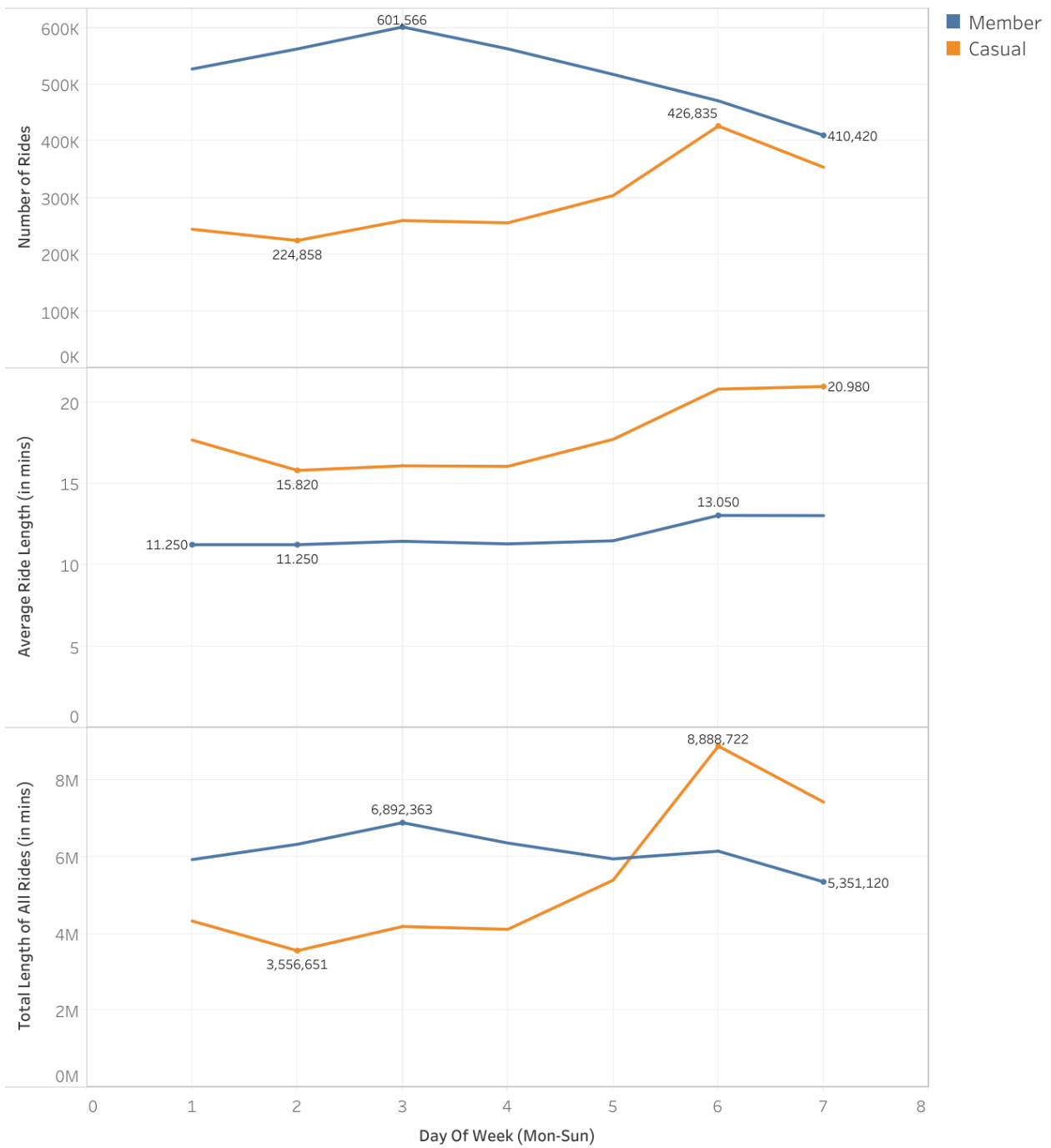
Summary - Whole of 2024



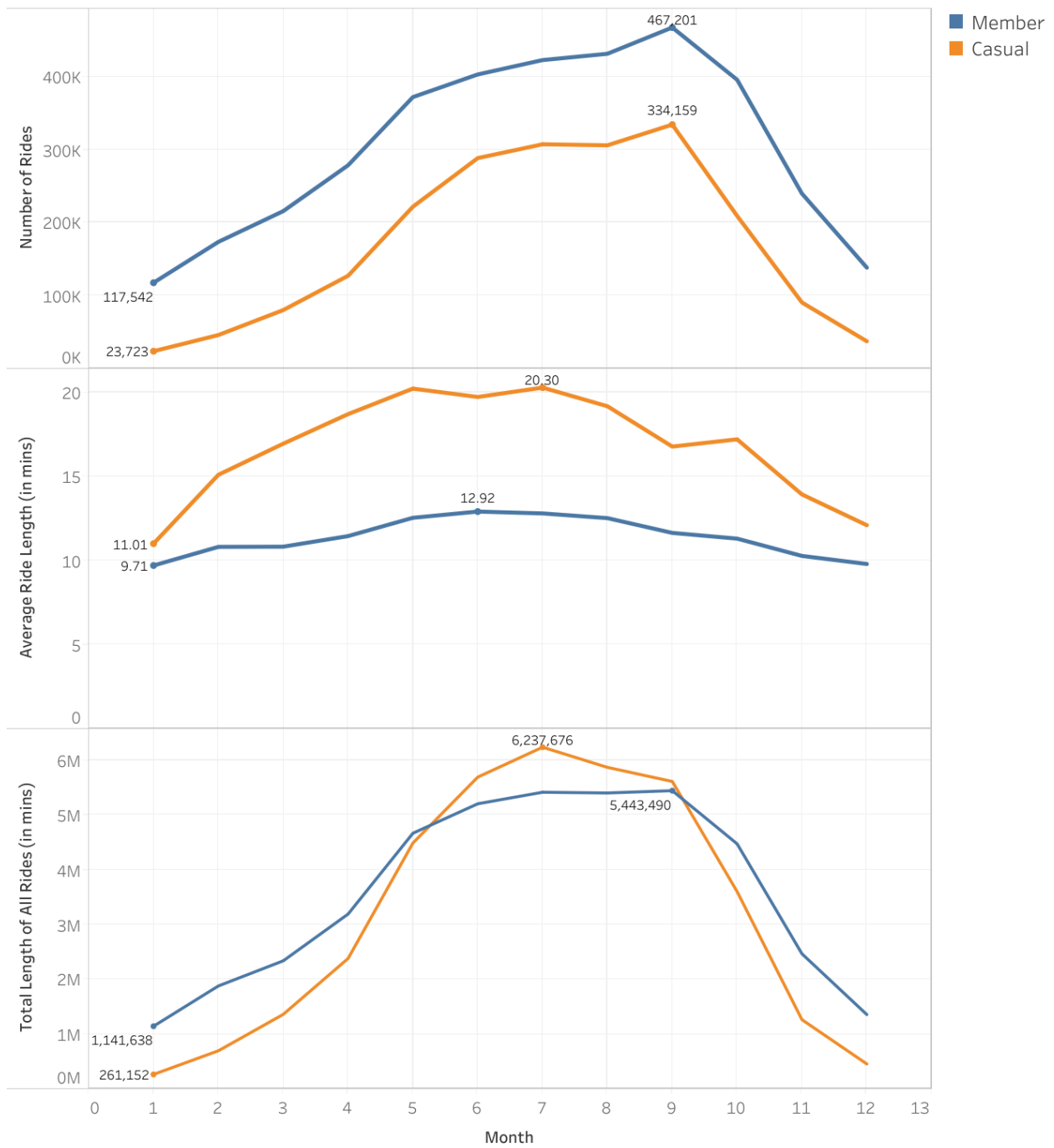
Breakdown by Hour-of-the-Day



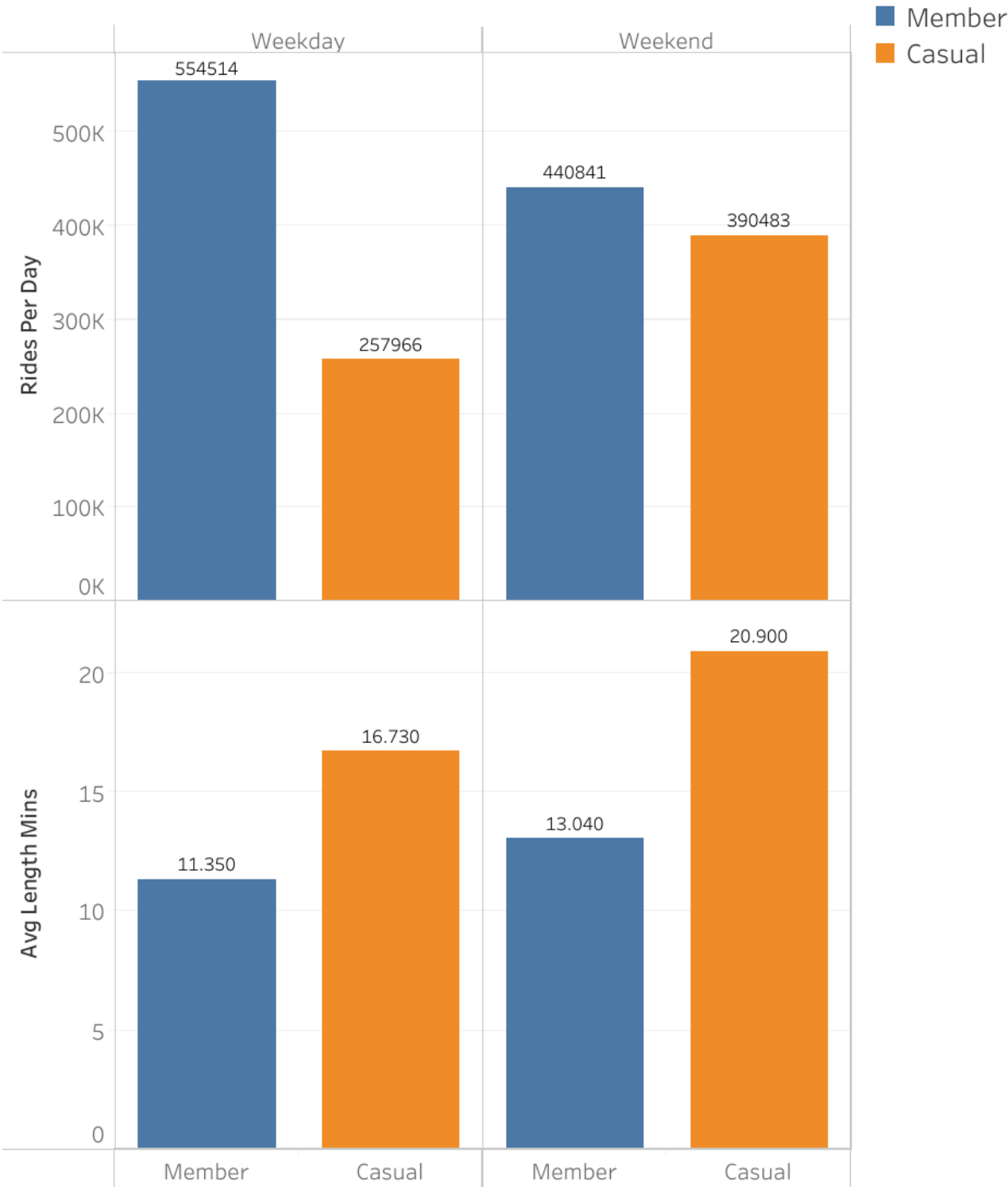
Breakdown by Day-of-the-Week



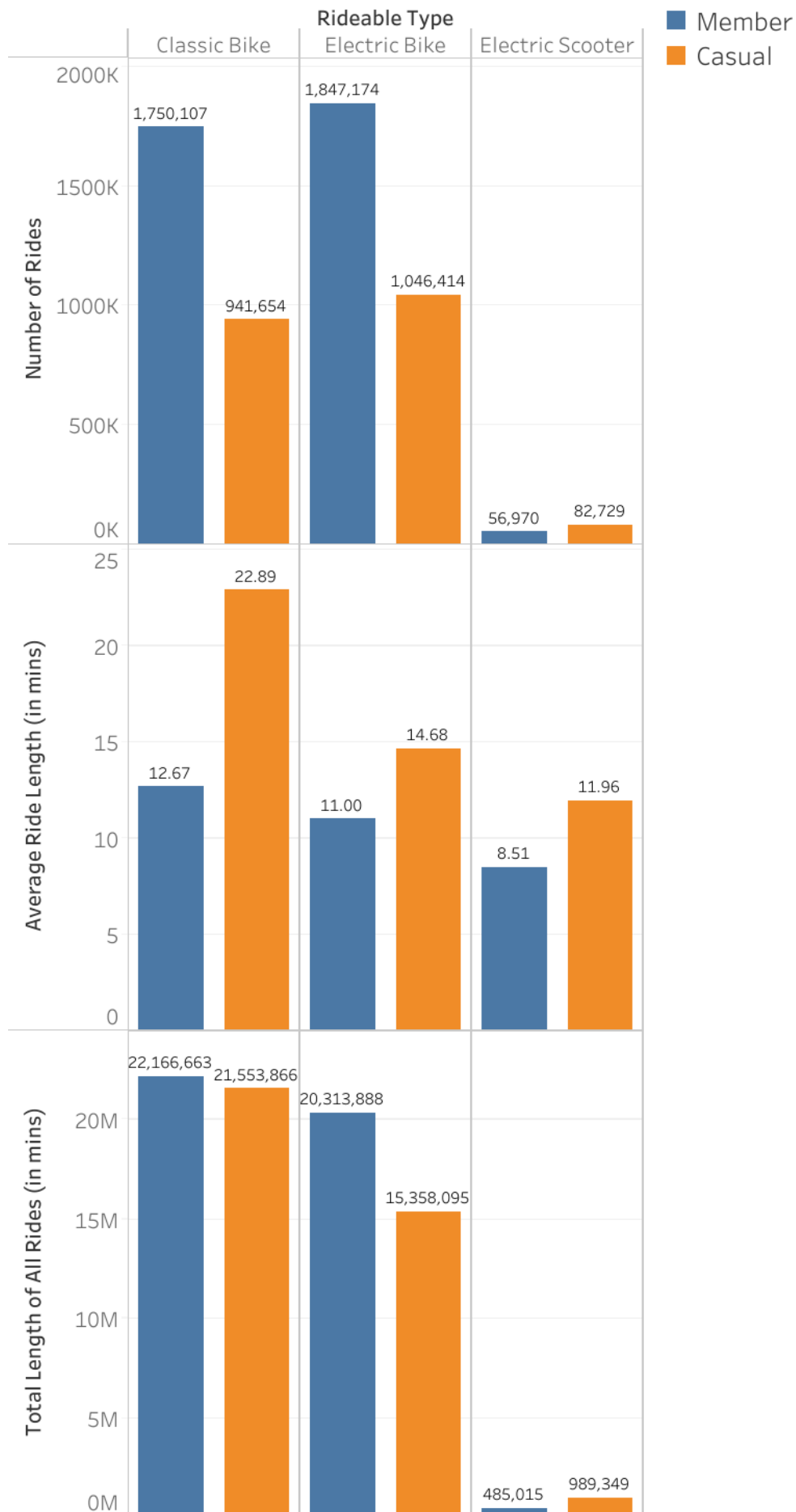
Breakdown by Month



Breakdown by Weekday / Weekend



Breakdown by Type-of-Rideable



Act

Report Recommendations

The data indicates distinct behavioural patterns between casual riders and members, suggesting different usage purposes, though with some overlap. Below I set out my recommendations to achieve the goals of the business task - converting Casuals to Members.

I recommend a targeted promotion scheduled in June, offering temporary discounts to casual riders if they start to use the service frequently during typical commuting hours of 8-9am and 3-6pm on weekdays, to encourage a long term behaviour change to using Cyclistic Bikes for commuting to work, or completing their commute from a railway station to their office. Encouraging commuting type journeys, also opens up the possibility of a return journey using Cyclistic. The June timing of the promotion, ensures access to a large pool of Casual users at that time, while still having the rest of the summer ahead to influence a comfortable behavioural change.

The promotion must require Casuals to sign up as Members, but the important part is the behavioural change we are influencing. Once the temporary promotion ends, these customers' habits will have changed and already having a Membership will mean it is convenient for them to continue their new habit; at which point our goal has been achieved.

Casual riders' existing revenue is a source of business that is essential to retain, in fact Casuals account for 13% more ridden minutes than Members. Therefore, I suggest that the above promotion rewards the customer in the form of discounts on their continued leisure-type usage. For example "Use Cyclistic bikes during weekday commuting hours, and get half price weekend rides." This ensures that Cyclistic does not lose the existing source of revenue by encouraging additional commuting use.

I would also recommend consideration of a further promotion, this time in collaboration with an outdoor clothing brand. This promotion should be scheduled in October, to encourage increased winter patronage on Cyclistic. Cyclistic is massively under-utilised during the winter months, likely due to cold and wet weather; quality outdoor clothing could mitigate this. This promo could take the form of customers being offered discounts on outdoor clothing from said outdoor clothing brand, on the condition that they keep their Cyclistic membership throughout the approaching colder months. The clothing discount would be funded by the clothing brand in return for access to our customer base, (the ideal target market for an outdoor clothing brand), maximising ROI for Cyclistic. This collaborative promotion will encourage continued Cyclistic Membership throughout the under-utilised months. The October timing coincides with a huge drop in patronage from the previous month, but the full winter season would still be ahead of us, so customers may be more willing to invest in outdoor clothing ahead of the cold and wet months.

I would also recommend Cyclistic's maintenance department are tasked with ensuring tyres on all rideables, especially classic bikes, have more-than-sufficient tread ahead of winter months, to ensure riding in the rain is as easy, fast and safe for customers as possible. This could be highlighted in the second promo campaign, to give less-confident riders peace of mind and more confidence; something which the data shows may separate less experienced Casual users and experienced members in wet October, on top of the seasonal trend.

I also recommend that promotions are introduced gradually, and monitored closely - Casuals complete more ridden minutes than Members; while our aim is to increase patronage, it is important to retain this revenue from current-Casuals even when they become Members. In the

event that the strategy is having the wrong effect, then a gradual introduction of the promotion will reduce losses, and an alternative strategy can then be devised.

If successful, this strategy will lead to a long term increase in company revenue.